Gina Marano

Forester, forest ecologist & forest modeller

Postdoctoral researcher, Swiss Federal Institute for Forest, Snow and Landscape Research WSL
Associate scientist, CNR-ISAFOM Forest Modelling Lab · Lecturer, ETH Zurich, D-USYS
Associate Editor, *Journal of Ecology*

Zürcherstrasse 111, 8903 Birmensdorf, Switzerland · gina.marano@wsl.ch · ginamarano.com
ORCID 0000-0003-2600-984X · Google Scholar · ResearchGate · LinkedIn

Profile

Forester and forest ecologist with a strong background in process-based modelling and silviculture. My work assesses how abiotic stress — above all drought and climate extremes — and biotic disturbances such as bark beetles and management interact and affect the growth, mortality and long-term dynamics of temperate forests. The goal is to (1) understand the processes leading to forest stress, (2) design management scenarios and simulations to assess their potential outcomes, and (3) support foresters with decisions grounded in both field and modelling expertise.

Positions

2025 – present
Postdoctoral Researcher (100%)
Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Birmensdorf, Switzerland
Research Unit Forest Resources and Management — Resource Analysis group, Swiss National Forest Inventory. Scientific lead for the further development of the MASSIMO management simulator in the project **ReForMASSIMO**: climate-sensitive growth, mortality and regeneration, a new disturbance module, and tests of when silvicultural interventions remain effective under climate change. Supervisors: Dr. Golo Stadelmann, Dr. Esther Thürig.

2026 – present
University Lecturer (33%)
ETH Zurich, Department of Environmental Systems Science (D-USYS), Switzerland
Master course *Quantitative Vegetation Dynamics*.

2018 – present
Associate Scientist
Forest Modelling Lab, CNR-ISAFOM, National Research Council of Italy
External research associate. Development of the management module and of the drought-mortality representation in the process-based model 3D-CMCC-FEM; applications on forest management, carbon sequestration and resilience.

2022 – 2023
Teaching Assistant & Research Support (20%)
ETH Zurich, D-USYS, Switzerland
Master courses *Quantitative Vegetation Dynamics* and *Gebirgswaldökologie und Management*.

2018 – 2020
Research Fellow (100%)
University of Naples Federico II, Italy
Horizon 2020 project **LANDSUPPORT**. Task group leader for the forestry tools in WP3: conception, development and deployment of regional and local forestry decision-support tools for the Campania Region (NUTS-2 and NUTS-3), with user training and stakeholder round tables.

Education

2021 – 2025

PhD in Forest Ecology

ETH Zurich, Switzerland

Thesis: *Unraveling the death spiral — a process-based assessment of the drivers and ecological responses of temperate forests to drought stress.* Supervisor: Prof. Dr. Harald Bugmann. SNSF grant 188882. doi:10.3929/ethz-c-000790347. Developed and validated new ForClim versions (v4.1–4.2) integrating predisposing, inciting and contributing stress factors; reproduced drought-induced mortality in Switzerland and Germany.

2015 – 2018

MSc Forest Sciences (110/110 cum laude)

University of Padua, Italy

Curriculum Forestry and Silviculture. Thesis: *Physico-chemical characterization of woody by-products for energetic purposes.*

2014 – 2015

Additional university coursework

University of Naples Federico II, Italy

GIS and Remote Sensing (30/30); Nature Conservation and Management of Protected Areas (30/30 cum laude); 18 ECTS recognised toward the MSc.

2011 – 2014

BSc Forest Sciences (110/110 cum laude)

University of Naples Federico II, Italy

Thesis: *Characterization of forest soils — a multiscale approach.*

2007 – 2011

Matura, European Classical Gymnasium (100/100)

Convitto Nazionale Vittorio Emanuele II, Naples, Italy

Silviculture & forestry practice

I regard models as instruments that aid silvicultural decisions by extending field experience beyond the range of past conditions: they make the long-term consequences of interventions visible, while expert knowledge tests their plausibility.

DECISION SUPPORT FOR PRACTITIONERS

2025 – present

MASSIMO — Swiss National Forest Inventory

WSL · for the Federal Office for the Environment (BAFU)

Climate-sensitive management scenario simulator for national forest development projections and the evaluation of management strategies mandated by federal policy. Improving growth, regeneration and mortality, the bark beetle disturbance module and the management modules.

2025

ForClim-App

ETH Zurich · PI Dr. Ulrike Hiltner

Provided the newest ForClim version with refined drought-induced mortality, integrated into the ForClim web application for projections of stand development.

2023 – 2024

ForClim-Marteloscope

ETH Zurich · teaching and practitioner training

Marteloscope tree-marking exercises linked to ForClim: silvicultural selection decisions are translated into model rules and confronted with projected growth, mortality, regeneration, timber, carbon and risk.

2023

Schweizerische Gebirgswaldpflegegruppe (GWG)

39th Arbeitstagung, Ilanz

Showed ForClim as decision support for Swiss mountain protection forests, in exchange with the mountain-silviculture practitioner group.

2018 – 2022

LANDSUPPORT S-DSS

University of Naples Federico II · H2020, GA 774234

Task group leader for the forestry tools: regional and local forestry tools and a machine-learning forest simulator supporting silvicultural planning in Italy and Slovenia.

2014 – 2016

SoilConsWeb

LIFE+ · CRISP, University of Naples Federico II

Forest plot surveys for a web-based spatial decision-support system for soil and landscape conservation (Camposauro Regional Park).

FIELD TRAINING & PROFESSIONAL FORESTRY

2019 – 2020

Forest-management planning support

Forestry Directorate, Campania Region, Italy · supervisor Alberto Mattia

2019

Forestry intern — forest-management planning

Matese Regional Park, Campania · supervisor Pio Ciliberti

2015

Forestry intern — tree marking and silvicultural planning

Campania · supervisor Alessandro Simeone

Field surveys, cartography, tree marking and planning of interventions in high-forest and coppice stands of *Castanea sativa* and *Corylus avellana*.

2013 – 2014

Forestry intern — site and stand assessment

Camposauro Regional Park, Telesina Valley · supervisor Simone Valentini

Dominant-height measurements and soil surveys in high-forest and coppice stands.

Forest operations	Chainsaw operations and safety, with final exam (CESFAM Paluzza, 2017); cable crane (2017), forest road network planning (2016) and forest operations (2016), BFW Forstliche Ausbildungsstätte Ossiach, Austria.
Ungulates & browsing	Monitoring and biometric censuses of fallow deer, wild boar and red deer (Castelporziano, 2018), of the endemic Mesola red deer (2015–2017) and of red deer on the Cansiglio and Asiago plateaus (2017), University of Padua.
Large carnivores	Marsican brown bear monitoring (Salviamo l'Orso, 2013–2014); wolf-damage recognition and mitigation course (Cilento National Park, 2014).

SCIENCE–PRACTICE TRANSFER

2021 – present	Public forest walks (*Waldumgänge*) on silviculture and forest change, with the district forester of Benken ZH; bridge between operational forestry and research there (ETH forest planning course, field testing of ForestStudio, stand selection for a dendroecological drought study).
2020	Contribution to the proposal of the COST Action on forest decision support, later funded as CA22141 *DSS4ES*.

Grants & projects

2026 – present

Swiss–New Zealand forest collaboration

Auckland University of Technology · seed funding NZD 5,000 · Role: co-developer

Exchange of forest data, models and adaptation approaches, testing which Central-European silvicultural principles are transferable. Host: Prof. Sebastian Leuzinger.

2025 – 2027

ReForMASSIMO

WSL · BAFU · CHF 500,000 · Role: postdoc, scientific lead for MASSIMO development

Climate-sensitive metamodel for growth, mortality and regeneration in all NFI plots; new disturbance module; plausibility of management strategies.

2021 – 2025

FORMULATE-DROUGHT — Embracing structural uncertainty in models of forest dynamics

ETH Zurich · SNSF grant 188882 · CHF 959,036 · Role: doctoral researcher and subproject contact

2018 – 2022

LANDSUPPORT

University of Naples Federico II · EU H2020 (774234) · € 6,999,771 total, € 1,351,583 institutional · Role: research fellow

Web-based land decision support linking silvicultural options with land-use objectives, stakeholder needs and multi-scale public decision-making.

2016 – 2017

PRO.PRO.FOR. — Promozione della Professione Forestale

University of Padua, Dept. TESAF · € 5,850 award · Role: project lead and funding recipient

Open seminar series and field summer school on silvicultural planning, forest operations, environmental assessment, monitoring and precision forestry.

Teaching

2026 – present

Quantitative Vegetation Dynamics: Models from Tree to Globe (MSc)

ETH Zurich, D-USYS · Lecturer

2022 – 2023

Forest gap modelling laboratory (MSc)

Quantitative Vegetation Dynamics, ETH Zurich · Teaching assistant (3-hour lab)

2022 – 2023

Gebirgswaldökologie und Management (MSc)

ETH Zurich · Laboratory and teaching material, with Prof. Harald Bugmann

2018 – 2019

Best practices in forest operations to minimise erosion and soil compaction

Guest lecture, BSc *Forest Pedology*, University of Naples Federico II

Also a crash course in R for geospatial analyses (MSc *Geography and Soil Evaluation*).

Supervision of theses

Master

2024–2025: Livia Gysel (ETH Zurich) — modelling support; main supervisor Dr. Mathieu Levesque.
2023: Francesca Eccher (Free University of Bolzano) — modelling support; main supervisor Dr. Marco Mina, Eurac Research.
2022–2023: Laurin Hillebrand (University of Innsbruck) — modelling support; main supervisor Dr. Marco Mina.
2021: Alessandro Aquino (University of Naples Federico II) — co-supervisor; main supervisor Prof. Fabio Terribile.

Bachelor

2024: Raphael Schmid (ETH Zurich) — co-supervisor; main supervisor Prof. Dr. Harald Bugmann.
2023: Lara Wüthrich (ETH Zurich) — modelling support; main supervisor Prof. Dr. Harald Bugmann.
2022: Samuel Zweifel (ETH Zurich) — modelling support; main supervisor Dr. Ulrike Hiltner.

Editorial & review service

Editorial roles

Associate Editor, *Journal of Ecology* (British Ecological Society). Editorial board member, *Current Climate Change Reports* (Springer). Guest Assistant Editor, *Forests* — Special Issue *Modeling Forest Response to Climate Change*.

Peer review

Nature Communications, New Phytologist (Tansley Review), Global Change Biology, Journal of Ecology, Geoscientific Model Development, Biogeosciences, Ecological Modelling, Scientific Reports, European Journal of Soil Science, Discover Sustainability, Discover Forests.

Institutions

Reviewer for the National Science Centre, Poland.

Professional affiliations

2026 – present

British Ecological Society, Forest Ecology Group — Scientific Advisor.

2026 – present

International Tree Mortality Network — Core team member, contributing expertise on dynamic vegetation forest models.

2020 – present

SISEF, Italian Society of Silviculture and Forest Ecology — Communication Officer of the Modelling Working Group.

Member

Schweizerischer Forstverein · SwissForestLab · Gesellschaft für Ökologie · Ecological Society of America

Student years	President, Forestry Student Association (AUSF / IFSA local committee), University of Padua (2017–2018); Vice-President, University of Naples Federico II (2012–2014); student representative in the academic councils of both forestry programmes.

Prizes & awards

2023	Best Oral Presentation Award, SwissForestLab Science Day
2020	Best Poster Award — Umberto Bagnaresi Prize, SISEF Congress of Silviculture
2011	Scholarship and selection as Best Student — Alfieri del Lavoro, President of the Italian Republic

Forest models — contributions

MASSIMO (Java)	Empirical, stochastic, individual-tree management simulator of the Swiss NFI. Enhancing the climate sensitivity of regeneration, growth and mortality through a climate meta-model; hybridisation with process-based approaches (ForClim).
ForClim v4.1–4.2 (C#, Python)	Gap model of stand dynamics. Main developer of versions 4.1 and 4.2, with enhanced sensitivity to climate extremes and drought-induced mortality. Ongoing development with ETH Zurich, University of Auckland and Universidad de Santiago de Chile. Open source.
3D-CMCC-FEM (C)	Process-based forest ecosystem model. Developer of the management module; applications on management, carbon sequestration and resilience. Open source.
ForestStudio (Rust)	Forest management model. Scientific support to development at the ETH Zurich Forest Resources Management professorship, including its silvicultural application with practitioners. Open source.

Selected talks

2025	**Modellierung zukünftiger Baumarteneignung: Herausforderungen und stetige Weiterentwicklungen** (invited) Montagskolloquium für die Praxis, ETH Zürich, 24 November 2025
2024	**Predisposing and inciting stress factors leading to drought-induced mortality** IUFRO World Congress, session T5.22, Stockholm, June 2024
2023	**Predisposing and inciting stress factors leading to drought-induced tree mortality** (best oral presentation) SwissForestLab Science Day, ETH Zurich, October 2023
2023	**From dynamic forest models to decision support for forest transformation — how can synergies be achieved?** (session chair) European Conference of Ecological Modelling, Leipzig, September 2023 Also presented *Predisposing and inciting stress factors leading to beech mortality in Switzerland* and a poster on dynamic forest models as decision support in Swiss mountain protection forests.
2022	**Effects of drought on Swiss forest dynamics — what is the influence of soil water storage capacity?** Forest Disturbances and Ecosystem Dynamics in a Changing World, Berchtesgaden National Park, Germany, September 2022
2022	**Analyzing the drought–growth relationship at six low-elevation beech sites in Switzerland** XIII Congresso Nazionale SISEF, Orvieto, Italy, May–June 2022
2021	**How to integrate dynamic forest models in geospatial decision support systems** LANDSUPPORT H2020 Round Table on Forestry, 25 October 2021

Publications — peer-reviewed

Bold indicates own name. † first or co-first author; ∗ corresponding author.

1. **Marano, G.**†∗, Hiltner, U., Meusburger, K., Hands, T.O., Bugmann, H. Predicting drought-induced tree mortality in Swiss beech forests hinges upon predisposing and inciting factors. *Ecosphere*, 2026. doi:10.1002/ecs2.70771
2. **Marano, G.**†∗, Hiltner, U., Knapp, N., Bugmann, H. Simulating the recent drought-induced mortality of European beech and Norway spruce in German forests. *Geoscientific Model Development*, 19(3), 1121–1141, 2026. doi:10.5194/gmd-19-1121-2026
3. **Marano, G.**†, Costa, M., Bugmann, H., Griess, V.C., Rudow, A., Díaz-Yáñez, O. Preparing for tomorrow: a protocol for tree species parameterization in dynamic vegetation models. *Ecological Modelling*, 517, 111587, 2026. doi:10.1016/j.ecolmodel.2026.111587
4. Boukhris, I., Cherubini, F., Collalti, A., Dalmonech, D., Vonderach, C., **Marano, G.** et al. Long-term forest-sector mitigation and radiative forcing under contrasting management, climate and substitution pathways. *Journal of Environmental Management*, 405, 129586, 2026. doi:10.1016/j.jenvman.2026.129586
5. Boukhris, I., **Marano, G.**†, Dalmonech, D., Valentini, R., Collalti, A. Modeling forest growth under current and future climate. *Current Forestry Reports*, 11(1), 17, 2025. doi:10.1007/s40725-025-00249-5
6. Fischer, R., Anders, T., Bugmann, H., Djahangard, M., Dreßler, G., Hetzer, J., Hickler, T., Hiltner, U., **Marano, G.** et al. Perspectives for forest modeling to improve the representation of drought-related tree mortality. *Journal für Kulturpflanzen*, 77(2), 50–69, 2025. doi:10.5073/JfK.2025.02.05
7. **Marano, G.**†, Dalmonech, D., Collalti, A. Modeling forest response to climate change (editorial). *Forests*, 15(7), 1194, 2024. doi:10.3390/f15071194
8. **Marano, G.**† et al. Weshalb sind die Buchen in Mittelland-Wäldern gestorben? *Schweizerische Zeitschrift für Forstwesen*, 175(1), 2024.
9. Dalmonech, D., Vangi, E., Chiesi, M., Chirici, G., Fibbi, L., Giannetti, F., **Marano, G.**, Massari, C. et al. Regional estimates of gross primary production applying the process-based model 3D-CMCC-FEM vs. remote-sensing multiple datasets. *European Journal of Remote Sensing*, 57(1), 2301657, 2024. doi:10.1080/22797254.2023.2301657
10. Terribile, F. et al., incl. **Marano, G.** The LANDSUPPORT geospatial decision support system (S-DSS) vision. *Land Degradation & Development*, 35(2), 813–834, 2024. doi:10.1002/ldr.4954
11. Vangi, E., Dalmonech, D., Cioccolo, E., **Marano, G.** et al. Stand age diversity (and more than climate change) affects forests' resilience and stability, although unevenly. *Journal of Environmental Management*, 366, 121822, 2024. doi:10.1016/j.jenvman.2024.121822
12. Testolin, R., Dalmonech, D., **Marano, G.** et al. Simulating diverse forest management options in a changing climate on a *Pinus nigra* subsp. *laricio* plantation in Southern Italy. *Science of the Total Environment*, 857, 159361, 2023. doi:10.1016/j.scitotenv.2022.159361
13. Dalmonech, D.†, **Marano, G.**†∗, Amthor, J.S., Cescatti, A., Lindner, M., Trotta, C., Collalti, A. Feasibility of enhancing carbon sequestration and stock capacity in temperate and boreal European forests via changes to management regimes. *Agricultural and Forest Meteorology*, 327, 109203, 2022. doi:10.1016/j.agrformet.2022.109203
14. Mahnken, M., Cailleret, M., Collalti, A. et al., incl. **Marano, G.** Accuracy, realism and general applicability of European forest models. *Global Change Biology*, 28(23), 6921–6943, 2022. doi:10.1111/gcb.16384
15. De Faria, B.L., **Marano, G.** et al. Model-based estimation of Amazonian forests recovery time after drought and fire events. *Forests*, 12(1), 8, 2021. doi:10.3390/f12010008
16. **Marano, G.**†, Collalti, A. "Models of tree and stand dynamics": a differential journey through forest modelling (book review). *iForest*, 13(1), 152–153, 2020. doi:10.3832/ifor0061-013
17. **Marano, G.**†, Langella, G., Basile, A., Cona, F. et al. A geospatial decision support system tool for supporting integrated forest knowledge at the landscape scale. *Forests*, 10(8), 690, 2019. doi:10.3390/f10080690

Publications — books, chapters & dissertation

1. **Marano, G.** Unraveling the death spiral: a process-based assessment of the drivers and ecological responses of temperate forests to drought stress. Doctoral dissertation, ETH Zurich, 2025. doi:10.3929/ethz-c-000790347
2. Dalmonech, D., **Marano, G.**, Collalti, A. (eds.) Modeling Forest Response to Climate Change. MDPI Books, 266 pp., 2024. doi:10.3390/books978-3-7258-1758-0
3. Collalti, A., Dalmonech, D., Vangi, E., **Marano, G.** et al. Monitoring and Predicting Forest Growth and Dynamics. CNR Edizioni, Roma, 2024. ISBN 978-88-8080-655-4
4. Mencarelli, A., Cavalli, R., **Marano, G.**, Povolo, M., Greco, R. Energy valorization of fruit shells and stones deriving from the food industry. In: *AIIA 2022*, Lecture Notes in Civil Engineering 337, Springer, 2023. doi:10.1007/978-3-031-30329-6_76

Publications — for practitioners

1 **Marano, G.**, Dalmonech, D. Aumento di temperatura e sequestro di carbonio: un punto di non ritorno a breve termine per le foreste? SISEF, March 2021.

2 **Marano, G.**, Castello, P. Dall'università alla professione. L'esperienza del progetto PRO.PRO.FOR. *Sherwood — Foreste ed Alberi Oggi*, 232, 2018.

3 **Marano, G.**, Zambon, G. Uniforest. Un'impresa familiare nel grande mercato europeo. *Tecniko & Pratiko*, 129, 2017.

Technical skills

Programming	C# (advanced); C, Python, Java, JavaScript, Rust (basic)
Scripting & stats	R; SAS, SPSS, Statgraphics Centurion, MATLAB
Modelling env.	RStudio, NetLogo, Jupyter Notebook, Google Colab
Stand visualisation	SVS — Stand Visualization System
Geospatial & CAD	QGIS, ArcGIS, Google Earth Engine; forest road network design
Development	Microsoft Visual Studio, Eclipse; Git (GitHub, GitLab)
HPC & cloud	ETH Euler cluster (SLURM, MPI/OpenMP)
Databases	PostgreSQL, Microsoft Access
Authoring	LaTeX, Quarto, Markdown, Shiny
Certificates	GIS, specialised level (AICA, 2015); ITsec4RI cybersecurity (ETH domain, 2026); data protection (British Ecological Society, 2026)

Languages & personal

Languages	Italian (native), English (C1), German (B2), Spanish (B2), French (A2), Norwegian (A1)
Personal	Born 15 March 1992 in Naples, Italy. Italian nationality. Based in the Canton of Zurich, Switzerland.